CivilSuite — User guide

Formulas, assumptions, limits and references · Version 1.1.0 (build 11) · English

1. Getting started

CivilSuite brings together seven site-engineering tools: concrete mix design, crack diagnosis, strength gain, rebar mass, concrete volumes, mortar and site reports. Each module shows its inputs, the results, the calculation steps, the checks and the warnings, and exports a PDF report.

- **Access.** The rebar module is free. The other modules and .civilsuite file sharing require Premium access: a monthly or yearly subscription (3-day free trial for new subscribers) or a one-time lifetime purchase. Manage it in Settings › Manage Premium access; “Restore purchases” recovers a purchase on a new device.
- **Units.** Settings › Units: SI or US customary. Calculations are always made in SI; only input and display change. In US units, only ACI and ASTM methods are available (Eurocode/EN methods are greyed out and stay readable). Reinforcement always stays metric.
- **Languages.** The app language (Settings › Language, applied at the next launch), the PDF report language (Settings) and the help language (menu at the top of this page) are set separately. The help exists in 14 languages; the PDF button shares the guide in the language shown.
- **Saving.** The bar at the bottom of each module saves the calculation under a title. Saved calculations, import and help are reached through the tray icon on the home screen. Recalculate any calculation saved with an older version before reusing its results.
- **iCloud.** Settings › Storage: calculations and reports can be kept in your own iCloud Drive to share them between your devices. A deletion is propagated to the other devices. If the iCloud account changes, only the data of the current account is shown.
- **Sharing.** PDF reports go through the iOS share sheet. A .civilsuite file (Premium) carries the selected calculations and reports with their photos; GPS positions are removed by default; its integrity is checked by a SHA-256 checksum. The declared author is not a signature.
- **Data safety.** If a saved calculation becomes unreadable, it is skipped without blocking the others and a copy of the original file is kept in quarantine in the app's private storage. A sound backup is made before every save; it is used for restoration if the whole file is unreadable. .civilsuite import is limited to 150 MB (200 MB uncompressed) and refused if available memory is insufficient.

IMPORTANT

Every result is a decision aid. It does not replace the applicable standards and their National Annexes, laboratory testing or the checks of a qualified engineer, who remains responsible for design and safety decisions.

2. Concrete mix design

Two methods are available: Eurocode (EN 206 with the Bolomey–Dreux relationship) and ACI (ACI 211.1 with the limits of ACI 318). Both use the absolute-volume method for 1 m³ of concrete. The result is a theoretical starting mix that must be confirmed by trial batches and specimens (EN 12390 / ASTM C39) before production.

Eurocode / EN 206 method

> $f_{cm} = f_{ck} + 8\ \text{MPa}$ (EN 1992-1-1, Table 3.1)
>
> $f_{cm} = f_{ck} + \max(8;\ 1.48 \cdot \sigma)$ (option)

Target mean strength from the characteristic cylinder strength f_{ck} (12 to 60 MPa). The statistical option uses the standard deviation σ of your production; it is a starting assumption, not a proof of conformity.

> $f_{cm} = G \cdot f_{ce} \cdot (C/W - 0.5) \Rightarrow W/C = 1 / [\ f_{cm} / (G \cdot f_{ce}) + 0.5\ ]$

G is the granular coefficient (Dreux, Table VII: 0.35 to 0.65 depending on aggregate quality and size; default 0.50) and f_{ce} the actual 28-day strength of the cement (EN 196-1). The app warns when G is outside 0.35–0.65. In Dreux's original method the water term is the total water on dry aggregates; the app applies the formula to effective water (EN 206), so G must be calibrated on trial batches with the same convention. Above C50/60 a water/binder ratio from trials is required.

Several exposure classes can be combined. The governing values are the lowest maximum w/c, the highest minimum strength class and the highest minimum binder content. Values recommended by EN 206 Annex F (informative, normal-weight concrete, 50-year working life); the National Annex prevails (NF EN 206/CN, BS 8500-1…). Additions (fly ash, slag…) are refused in EN mode without a validated k-value.

EN 206 Table F.1 — recommended limiting values

Class	max w/c	min. strength class	min. binder (kg/m³)	air
X0	—	C12/15	—	—
XC1	0.65	C20/25	260	—
XC2	0.60	C25/30	280	—
XC3	0.55	C30/37	280	—
XC4	0.50	C30/37	300	—
XS1	0.50	C30/37	300	—
XS2	0.45	C35/45	320	—
XS3	0.45	C35/45	340	—
XD1	0.55	C30/37	300	—
XD2	0.55	C30/37	300	—
XD3	0.45	C35/45	320	—
XF1	0.55	C30/37	300	—
XF2	0.55	C25/30	300	≥ 4.0 %
XF3	0.50	C30/37	320	≥ 4.0 %
XF4	0.45	C30/37	340	≥ 4.0 %
XA1	0.55	C30/37	300	—
XA2	0.50	C30/37	320	—
XA3	0.45	C35/45	360	—

Limiting values: French National Annex (NF)

“Limiting values” setting: CEN recommended values (EN 206, Annex F) or the **French National Annex**. The French annex is the only one built in: its values could be checked against public sources (NF EN 206-1 of April 2004, Table NA.F.1, and the equivalences of §NA.4.1: XC2 = XC1; XC3, XC4 and XD1 = XF1; XS1 = XS2). The British annex (BS 8500-1) is not built in, for lack of verifiable access to all of its tables. The NF EN 206/CN (2014) and NF EN 206+A2/CN (2022) editions may change some values (for example the minimum binder in XA3): check against your copy in force. French rules on additions (k-value), cement types and FD P 18-011 are not modelled.

NF EN 206-1 (2004) Table NA.F.1 — French limiting values

Class	max w/c	min. strength class	min. binder (kg/m³)	air
X0	—	—	150	—
XC1, XC2	0.65	C20/25	260	—
XC3, XC4, XD1, XF1	0.60	C25/30	280	—
XF2	0.55	C25/30	300	≥ 4.0 %
XF3	0.55	C25/30	315	≥ 4.0 %
XF4	0.45	C30/37	340	≥ 4.0 %
XS1, XS2, XD2, XA1	0.55	C30/37	330	—
XS3, XD3, XA2	0.50	C35/45	350	—
XA3	0.45	C40/50	385	—

EN 206 gives no water-demand method: the initial effective water comes from ACI 211.1 Table A1.5.3.3 for the slump and the nominal maximum aggregate size (constant within the bands 25–50, 75–100 and 150–175 mm, interpolated between bands). Enter the certified nominal size; otherwise the D90 of the coarse grading is used and flagged. Water measured on trial batches can replace the tabulated water.

$$P(d) = A + (100 - A) \cdot \sqrt{(d / D)} \quad (d \le D)$$

$$\min \Sigma \left[P_{blend}(d_i) - P(d_i) \right]^2$$

Aggregate proportions: automatic least-squares fit of the blend to the Bolomey reference curve (A from 0 to 30; A = 0 gives Fuller), up to 8 aggregates, or manual percentages. Gradings must have distinct sieves and cumulative passing values increasing from 0 to 100 %.

Dreux reference curve and water convention

“Reference curve” setting: Bolomey (default) or Dreux's OAB broken line (Techniques de l'Ingénieur C2220, 1982, §2.64). O is at 0 % for 0.08 mm, B at 100 % for D; the line is straight on the logarithmic axis of the grading chart. K comes from Table IX below, K_s from the fineness modulus of the first sand (AFNOR sieves 0.16 to 5 mm used by Dreux) and K_p (+5 to +10) applies to pumpable concrete. As in the original method, the resulting proportions are absolute volumes of aggregate; with Bolomey they are mass fractions.

$$X = D/2 \ \ (D \le 20\ \text{mm});\ X = \sqrt{(5 \cdot D)} \ \ (D > 20\ \text{mm})$$

$$Y = 50 - \sqrt{D} + K + K_s + K_p \ \ (+ (D - 25)/5 \ \ \text{if } D > 50\ \text{mm});\ K_s = 6 \cdot M_f - 15;\ M_f = \Sigma\, R(0.16 \ldots 5\ \text{mm}) / 100$$

$$P(d) = Y \cdot \ln(d/0.08) / \ln(X/0.08) \ \ (0.08 \le d \le X);\ P(d) = Y + (100 - Y) \cdot \ln(d/X) / \ln(D/X) \ \ (X \le d \le D)$$

Dreux, Table IX — coefficient K

Cement content (kg/m³)	Vibration low · rounded	Vibration low · crushed	Vibration normal · rounded	Vibration normal · crushed	Vibration strong · rounded	Vibration strong · crushed
400 + plasticiser	−2	0	−4	−2	−6	−4
400	0	+2	−2	0	−4	−2
350	+2	+4	0	+2	−2	0
300	+4	+6	+2	+4	0	+2
250	+6	+8	+4	+6	+2	+4
200	+8	+10	+6	+8	+4	+6

“Water convention (Bolomey)” setting: by default the Bolomey relation is applied to effective water (EN 206). With “Total water on dry aggregates (Dreux)”, C/W applies to total water, i.e. effective water plus the water absorbed by dry aggregates; the cement content is solved exactly, because the aggregate volume itself depends on the cement. Durability limits (max w/c, min binder) are still checked on effective water. Calibrate G with trial batches using the same convention.

$$W_{tot} = W + \Sigma\, M_i \cdot abs_i/(1 + abs_i);\ C = (C/W) \cdot W_{tot}$$

$$\alpha = \Sigma\, m_i \cdot abs_i/(1 + abs_i);\ V_0 = 1000 - W - 10 \cdot air$$

$$C = (C/W) \cdot (W + \alpha \cdot V_0) / [\ 1 + (C/W) \cdot \alpha \cdot 1000 / \rho_c\ ];\ (W/C)_{eff} = W / C$$

$$V_{agg} = 1000 - W - 1000 \cdot C/\rho_c - 10 \cdot air \ \ (L/m^3)$$

$$M_{SSD} = V_{agg} / [\ 1000 \cdot \Sigma\, (w_i / \rho_i)\ ]$$

$$M_{dry} = M_{SSD} / (1 + abs/100)$$

$$M_{batch} = M_{dry} \cdot (1 + w/100)$$

$$W_{add} = W - \Sigma\, (M_{batch} - M_{SSD})$$

Particle densities are SSD; absorption and moisture are percentages of dry mass. A negative water to add (wet aggregates) invalidates the mix. Admixture water and wash water must be accounted for separately.

WORKED EXAMPLE

f_{ck} = 30 MPa · XC1 · slump 100 mm · D = 16 mm (column 19 mm) · G = 0.50 · f_{ce} = 42.5 MPa

f_{cm} = 38 MPa; C/W = 2.288; W/C = 0.437; W = 205 L; air = 2 %; C = 469 kg

sand 57 % / gravel 43 %; M_{SSD} = 953 + 719 kg; W_{add} = 185 L; ρ = 2347 kg/m³

ACI 211.1 / ACI 318 method

$f'_c < 21$ MPa: $f'_{cr} = f'_c + 7.0$

$21 \le f'_c \le 35$ MPa: $f'_{cr} = f'_c + 8.3$

$f'_c > 35$ MPa: $f'_{cr} = 1.10\, f'_c + 5.0$

$f'_c \le 35$ MPa: $f'_{cr} = \max(\, f'_c + 1.34\, s;\ f'_c + 2.33\, s - 3.5\,)$

$f'_c > 35$ MPa: $f'_{cr} = \max(\, f'_c + 1.34\, s;\ 0.90\, f'_c + 2.33\, s\,)$

$s = k \cdot s_n$; k = 1.16 (n = 15) · 1.08 (20) · 1.03 (25) · 1.00 (≥ 30), linear interpolation

Required average strength f'_{cr} per ACI 301-16 §4.2.3.3 (referenced by ACI 318-19 §26.4.3.1(b)). The standard deviation is used only with at least 15 documented tests; the modification factor k of Table 4.2.3.3.a is linearly interpolated between 15 and 30 tests. In US units the native psi values are used: +1000 psi below 3000 psi, +1200 psi up to 5000 psi, 1.10 f'_c + 700 psi above, and −500 psi in the formula with s.

The w/c ratio comes from ACI 211.1 Table A1.5.3.4(a) (15 to 40 MPa without entrained air, 15 to 35 MPa with entrained air), then the lowest value with the ACI 318-19 Table 19.3.2.1 limits of all selected exposure classes is kept. Freezing classes F1 to F3 require entrained air (ACI 318-19 Table 19.3.3.1). Class S3 is not modelled and is refused.

ACI 318-19 Table 19.3.2.1 — requirements by exposure class

Class	max w/cm	min f′c, MPa (psi)
F0 · S0 · W0 · C0	—	17 (2500)
F1	0.55	24 (3500)
F2	0.45	31 (4500)
F3	0.40	35 (5000)
S1	0.50	28 (4000)
S2	0.45	31 (4500)
S3	—	—
W1	—	17 (2500)
W2	0.50	28 (4000)
C1	—	17 (2500)
C2	0.40	35 (5000)

ACI 318-19 Table 19.3.3.1 — target total air content (%)

nominal max. aggregate (mm)	F1	F2 · F3
9.5	6.0	7.5
12.5	5.5	7.0
19	5.0	6.0
25	4.5	6.0
37.5	4.5	5.5
50	4.0	5.0
75	3.5	4.5

$$M_{coarse,dry} = (b/b_0) \cdot \rho_{rodded};\ M_{coarse,SSD} = M_{coarse,dry} \cdot (1 + abs/100)$$
$$V_{sand} = 1000 - (W + V_C + V_{SCM} + 10 \cdot air + V_{coarse})$$

Coarse aggregate: dry-rodded volume b/b_0 from ACI 211.1 Table A1.5.3.6 (sand fineness modulus 2.40 to 3.00, ASTM C125). Sand: the remaining absolute volume. Only the first sand and the first coarse aggregate are used.

- Outside the ACI strength table, a water/binder ratio from trial batches is required. With more air than its assumptions (2 % without, 6 % with air entrainment), the calculation is refused unless the option “Correct f′cr for excess air” is enabled (see ACI options).

- ACI additions require a water/binder ratio validated by trial batches.
- Cement types, chlorides, sulfates, alkali–aggregate reaction, cover and spacing of reinforcement are to be checked separately.
- A failed durability check invalidates the result; the warnings must be read before any export.
- The identified editions (ACI 211.1-91, EN 206:2013) are those of the implemented tables; check the editions that apply contractually.

ACI options

Excess air. The option “Correct f′cr for excess air” replaces the refusal by an equivalent strength (formula below), using the empirical rule of about 5 % strength loss per 1 % of air (ACI FAQ), which can be edited. It lies outside the scope of the ACI 211.1 table and must be calibrated by tests; above a 50 % correction the calculation is refused. **Plain concrete (F3).** Note [3] of ACI 318-19 Table 19.3.2.1: w/cm ≤ 0.45 and f'_c ≥ 31 MPa (4500 psi) instead of 0.40 and 35 MPa. **Seawater (S1).** Note [5]: portland cement with C_3A ≤ 10 % is permitted if w/cm ≤ 0.40; the C_3A content must be justified.

> $f'_{cr,eq} = f'_{cr} / [\ 1 - c \cdot (a - a_0) / 100\ ]$; a_0 = 2 % (without air entrainment) · 6 % (with air entrainment); c = 5.0 %/% (default)

3. Strength gain and maturity

Estimates the mean compressive strength at age J from a model curve, corrected for temperature by an equivalent age and, if available, calibrated on test results.

> $f_{cm}(t) = \beta_{cc}(t) \cdot f_{cm,28}$; $\beta_{cc}(t) = \exp\{\ s \cdot [\ 1 - (28 / t)^{1/2}\ ]\ \}$
> s = 0.20 (R) · 0.25 (N) · 0.38 (S)

> $f(t) = f_{28} \cdot t / (a + \beta \cdot t)$; a = 4.0; β = 0.85

EN 1992-1-1 (3.2): s depends on the cement class (R: CEM 42.5 R, 52.5 N/R; N: CEM 32.5 R, 42.5 N; S: CEM 32.5 N); $f_{cm,28} = f_{ck}$ + 8 MPa by default. ACI 209R-92: moist-cured concrete with type I cement; f_{28} defaults to the f'_{cr} of ACI 301 without production data.

> Nurse–Saul: $t_e = \int \max(T - T_0; 0) / (T_r - T_0)\ dt$
> Arrhenius: $t_e = \int \exp[\ -E_a/R \cdot (\ 1/(273.15 + T) - 1/(273.15 + T_r)\ )\ ]\ dt$
> EN 1992-1-1 (B.10): $t_T = \int \exp[\ 13.65 - 4000 / (273 + T)\ ]\ dt$

Equivalent age at the reference temperature T_r (ASTM C1074): 20 °C by default (EN), 23 °C (ACI), adjustable for Nurse–Saul and Arrhenius; B.10 keeps 20 °C. Defaults: E_a = 41.5 kJ/mol, T_0 = 0 °C, R = 8.314 J/(mol·K); to be determined for the actual mix. Temperatures are interpolated

linearly between readings; the first and last readings are extended. Example: a linear rise from 20 to 40 °C over 24 h gives t_e = 1.830 d (Arrhenius, T_r = 20 °C).

$$f_{28} = \Sigma\,(f_j \cdot r_j) \,/\, \Sigma\, r_j^2$$

Calibration on tests: the scale f_{28} is fitted by least squares and, with at least two distinct ages, the shape parameter (s or a) by a grid search. Each specimen has its own equivalent age (by default its real age at the reference temperature): the site temperature history is never applied to laboratory specimens. With one age only, only the scale is fitted.

Results: equivalent age and strength at J (prior and calibrated curves), table from 1 to 90 days, and equivalent age needed to reach a required strength (search up to 365 days).

IMPORTANT

Indicative estimate of the mean strength: it authorises neither striking, lifting nor loading. Conformity is judged at the age specified for the project (EN 206 / ACI 318). At three days or less, a reliable value requires tests.

4. Rebar, mesh and laps

$$m = 7850 \cdot \pi \cdot d^2 / 4 \;\; (\text{kg/m; d [m]}) \;\rightarrow\; \text{Ø12: } 0.888 \text{ kg/m}$$

$$m_{mesh} = m_{wire} \cdot (1/s_x + 1/s_y) \;\; (\text{kg/m²; s [m]})$$

Nominal diameters: EN 10080 / BS 4449 (Ø6 to Ø50), ABNT NBR 7480 (Ø5 to Ø40) and ASTM A615 (#3 to #18). Masses are geometric (ρ = 7850 kg/m³) and may differ slightly from tabulated nominal masses; mesh mass is an area average without edge wires, offcuts or laps. BS 8666:2020 shape codes are drawn for illustration: cutting length and minimum bending radius are not calculated.

Lap of welded mesh — EN 1992-1-1 §8.7.5

Main wires (ribbed wires only):

$$f_{ctd} = \alpha_{ct} \cdot f_{ctk,0.05} / \gamma_C \;(3.16);\; f_{bd} = 2.25 \cdot \eta_1 \cdot \eta_2 \cdot f_{ctd} \;(8.2)$$

$$l_{b,rqd} = (\varphi / 4) \cdot (\sigma_{sd} / f_{bd}) \;(8.3)$$

$$l_0 = \alpha_1\,\alpha_2\,\alpha_3\,\alpha_5\,\alpha_6 \cdot l_{b,rqd} \geq l_{0,min} \;(8.10)$$

$$l_{0,min} > \max(\, 0.3 \cdot \alpha_6 \cdot l_{b,rqd};\; 15\,\varphi;\; 200 \text{ mm} \,) \;(8.11)$$

$$\alpha_2 = 1 - 0.15\,(c_d - \varphi)/\varphi \in [0.7;\, 1];\; \alpha_5 = 1 - 0.04\,p \in [0.7;\, 1];\; \alpha_2 \cdot \alpha_3 \cdot \alpha_5 \geq 0.7$$

$$\alpha_6 = (\rho_1 / 25)^{0.5} \in [1.0;\, 1.5]$$

Recommended values γ_C = 1.5, γ_S = 1.15, α_{ct} = 1.0 (the National Annex prevails); $f_{ctk,0.05}$ from Table 3.1, capped at C60/75 (§8.4.2(2)); $\sigma_{sd} = f_{yd}$ by default (conservative); α_3 = 1; twin wires: $\varphi_n = \varphi\sqrt{2}$ and α_2 = 1. If the clear gap between lapped bars exceeds min(4φ; 50 mm), it is added

to the length. Lappable percentage: layered sheets 100 % if $(A_s/s)_{prov} \le 1200$ mm²/m, otherwise 60 %, joints staggered by 1.3 l_0; intermeshed sheets 100 % with one layer, 50 % with several. The length provided is rounded up to the next millimetre and strictly exceeds $l_{0,min}$. Secondary wires (Table 8.4): the lap must enclose two main wires; the app adds the offset e of the first wire and the full wire width.

EN 1992-1-1 Table 8.4 — laps of secondary wires

secondary wire Ø (mm)	min. lap (mm)	min. pitches in the lap
φ ≤ 6	150	1
6 < φ ≤ 8.5	250	2
8.5 < φ ≤ 12	350	2

WORKED EXAMPLE

Ø6 / 150 · C30/37 · f_{yk} = 500 MPa · γ_S = 1.15 · γ_C = 1.5

f_{bd} = 3.00 MPa; $l_{b,rqd}$ = 217.4 mm; l_0 = 0.7 × 1.5 × 217.4 = 228.3 mm → **229 mm**; 1.3 l_0 → **298 mm**

$l_{0,sec}$ = max(150; 0 + 1 × 150 + 6) = **156 mm**

IMPORTANT

This is not a full ULS/SLS section check. Check the drawing (clear distance between adjacent laps ≥ max(2φ; 20 mm), longitudinal offset ≥ 0.3 l_0), the bars actually present and the National Annex. No ACI 318 lap rule is implemented.

5. Concrete volumes

Duct bank, manhole and U-ditch: theoretical geometric volumes, without waste, chamfers, over-thickness or bulking.

$$B = c_l + c_r + n\cdot Ø + (n-1)\cdot e_h;\ H = c_t + c_b + m\cdot Ø + (m-1)\cdot e_v$$

$$V = B \cdot H \cdot L - N \cdot \pi \cdot Ø^2 / 4 \cdot L$$

$$V = A_{ext}\cdot e_{base} + (A_{ext}\cdot H - A_{int}\cdot H_{int} - A_{ext}\cdot e_{base} - A_{ext}\cdot e_{top}) + (A_{ext} - A_{opening})\cdot e_{top} - \Sigma\, V_{openings}$$

$$Ø_{res} = Ø_{ext} + 2\cdot j;\ z_{axis} = IL - e_{pipe} + Ø_{ext}/2$$

Manhole: base slab, walls and cover slab with its opening, minus wall openings and the inlet/outlet pipe sleeves. Levels: inside floor = top level − H + base thickness; IL = pipe invert level. Openings higher than the clear wall height or larger than the slab are refused. Without coordinates, overlaps and edge distances between openings are not checked; on a circular wall, area × thickness is an approximation.

$$S = (B_i + 2e) \cdot (H_i + e_b) - B_i \cdot H_i;\ V = S \cdot L$$

$$W = \gamma \cdot V \quad (\gamma = 25\ \text{kN/m}^3,\ 10\text{–}40);\ m = 1000 \cdot W / 9.81\ \text{kg}$$

$$0.200\ \text{m}^3 \rightarrow 5.00\ \text{kN} \rightarrow 509.7\ \text{kg}$$

Self-weight with γ = 25 kN/m³ for normal-weight reinforced concrete (EN 1991-1-1, Table A.1), adjustable from 10 to 40 kN/m³; the same physical value is used in SI and US units. The theoretical mass does not replace a lifting study (accessories, tolerances, water, dynamic effects).

6. Mortar

ASTM C270 cement–lime types (volume proportions) or a free cement–sand mix. Sand: 2.25 to 3 times the sum of the binder volumes. Reference bulk densities: cement 94 lb/ft³ (1506 kg/m³), hydrated lime 40 lb/ft³ (641 kg/m³), dry sand 80 lb/ft³ (1280 kg/m³); absolute densities give the yield.

ASTM C270 — proportion types (cement = 1)

Type	hydrated lime (parts)	min. 28-day strength (property spec.)
M	0.25	17.2 MPa (2500 psi)
S	0.25 < L ≤ 0.50	12.4 MPa (1800 psi)
N	0.50 < L ≤ 1.25	5.2 MPa (750 psi)
O	1.25 < L ≤ 2.50	2.4 MPa (350 psi)

$$V = [\ M_c/\rho_c + M_l/\rho_l + M_s/\rho_s + W/1000\ ] / (1 - \text{air});\ k = 1 / V$$

$$Q = k \cdot M \cdot V_{req} \cdot (1 + \text{waste}/100)$$

WORKED EXAMPLE

1: 0.5: 4.5 · W/B = 0.9 · air 2 % → 333 kg cement · 71 kg lime · 1273 kg sand · 363 L water / m³

IMPORTANT

The water/binder ratio is a starting value to adjust to the consistency. Quantities are for dry sand: measure moisture, absorption and bulking before batching. A proportion type does not prove the strength of the property specification; an EN 998-2 class (M1 … M20) must be shown by tests (EN 1015-11).

7. Crack diagnosis

Photograph the crack face-on with a scale in the same plane: a QR code of known side or a reference length. Draw a transect across the crack, from sound concrete to sound concrete.

$w = FWHM_{px} \cdot scale$ (mm/px); scale = QR side (mm) / mean QR side (px)

transect oblique by θ: $w_{meas} \approx w / \cos\theta$

Width = full width at half depth (FWHM) of the dark trough of the intensity profile, ±1 pixel. Internal reliability thresholds: 3 px and 15 % contrast. If the QR sides differ by more than 5 %, the camera was not perpendicular. The measurement is optical (light, texture, shadows); a crack-width gauge remains the reference.

Transect obliquity: the local crack orientation is estimated with the structure tensor (Sobel gradients around the intensity dip). The width read along a transect inclined by θ to the normal is w/cos θ; the app corrects it by cos θ up to 60° and shows the angle. Beyond 60° the measurement is refused (redraw the transect); if the orientation is undetermined (coherence < 0.5) the width is not corrected. These thresholds are internal to the app, not normative.

$w = w_{transect} \cdot \cos\theta \quad (\theta \le 60°)$

n: dominant eigenvector of $J = \Sigma \begin{bmatrix} g_x^2 & g_x g_y \\ g_x g_y & g_y^2 \end{bmatrix}$; coherence $= (\lambda_1 - \lambda_2) / (\lambda_1 + \lambda_2) \ge 0.5$

EN 1992-1-1 Table 7.1N — w_{max} (mm)

Exposure	w_{max} (mm)
X0, XC1	0.4
XC2, XC3, XC4	0.3
XD1, XD2, XS1, XS2, XS3	0.3
XD3	—

“Limiting values” setting (Eurocode): recommended values (Table 7.1N) or the **French National Annex** NF EN 1992-1-1/NA, Table 7.1NF (reinforced concrete and unbonded prestressing, quasi-permanent combination). Notes 2 and 3 of the table: in X0/XC1, and in XC2–XC4 for buildings of categories A to D, crack control is deemed satisfied by the detailing rules; XA classes are governed by the project contract documents. The British annex is not built in.

NF EN 1992-1-1/NA Table 7.1NF — w_{max} (mm), reinforced concrete

Exposure	w_{max} (mm)
X0, XC1	0.40
XC2, XC3, XC4	0.30
XD1, XD2, XD3, XS1, XS2, XS3	0.20

ACI 224R-01 Table 4.1 — reasonable crack widths

Exposure condition	Width
Dry air or protective membrane	0.41 mm · 0.016 in
Humidity, moist air, soil	0.30 mm · 0.012 in
Deicing chemicals	0.18 mm · 0.007 in
Seawater and seawater spray, wetting and drying	0.15 mm · 0.006 in
Water-retaining structures (excluding non-pressure pipes)	0.10 mm · 0.004 in

Classification: below the limit; above the limit (monitor or treat); more than twice the limit (internal rule: ask a structural engineer). The width is never rounded before classification. EN values (Table 7.1N, reinforced concrete or unbonded prestress, quasi-permanent combination) are recommended values; XD3 and bonded prestress are outside the classification. ACI 224R values are indicative “reasonable” widths under service loads.

Possible causes are ranked by a compatibility score (pattern 30, age 25, depth 10, element 10, activity 10, visual signs 15 points), not by a probability, following ACI 224.1R-07 and Concrete Society TR22.

IMPORTANT

An observed width is not a calculated w_k and a small width alone does not demonstrate safety. Diagonal cracks or visible deflection: ask a structural engineer without delay.

8. Site reports, weather and Heat Index

Calendar of reports: several reports per day are possible (place, trade, shift). Day and night shift hours are set in Settings (defaults 06:30–17:30 and 17:30–04:30 the next day) and frozen in each report, in the time zone of the site. A report gathers activities, quantities, personnel, equipment, photos (camera or library, position optional) and exports to PDF.

Weather comes from Apple Weather (WeatherKit) for the site position and the shift window (±1 h to interpolate the boundaries); values can also be entered by hand. The Heat Index is computed by CivilSuite from these data (modified data) with the NWS method:

$$HI_s = 0.5 \cdot [\, T + 61 + 1.2\,(T - 68) + 0.094\,RH \,];\ (HI_s + T)/2 < 80\ °F \rightarrow HI = (HI_s + T)/2$$

$$HI = -42.379 + 2.04901523\,T + 10.14333127\,RH - 0.22475541\,T{\cdot}RH - 0.00683783\,T^2 - 0.05481717\,RH^2 + 0.00122874\,T^2{\cdot}RH + 0.00085282\,T{\cdot}RH^2 - 0.00000199\,T^2{\cdot}RH^2 \quad (T\ [°F],\ RH\ [\%])$$

$$RH < 13\ \%\ ,\ 80 \le T \le 112\ °F:\ HI - [(13 - RH)/4] \cdot \sqrt{\{[17 - |T - 95|]\ /\ 17\}}$$

$$RH > 85\ \%\ ,\ 80 \le T \le 87\ °F:\ HI + [(RH - 85)/10] \cdot [(87 - T)/5]$$

$$70\ °F\ /\ 50\ \% \rightarrow 69.5\ °F \cdot 90\ °F\ /\ 70\ \% \rightarrow 105.9\ °F \cdot 96\ °F\ /\ 65\ \% \rightarrow 121.0\ °F$$

NWS Heat Index categories

Heat Index (°F)	Category
< 80	Out of range
80–90	Caution
91–103	Extreme caution
104–124	Danger
≥ 125	Extreme danger

$$HI_{mean} = \Sigma \left[(HI_k + HI_{k+1})/2 \cdot \Delta t_k \right] / \Sigma \Delta t_k$$

Shift statistics: maximum (including interpolated boundaries) and time of peak, time-weighted mean over the covered segments; gaps longer than 90 minutes are neither integrated nor interpolated. Categories are read on the value rounded to the nearest °F.

IMPORTANT

Heat Index in the shade with light wind, for information: full sun can add up to 15 °F (NWS). It is neither a WBGT nor a work-stoppage threshold; apply your site's heat-stress procedure.

9. References

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2. CEN — EN 206:2013, Concrete — Specification, performance, production and conformity. Annex F (informative), Table F.1.
3. CEN — EN 1991-1-1:2002, Eurocode 1: Actions on structures — Part 1-1. Annex A, Table A.1.
4. ACI Committee 211 — ACI 211.1-91 (Reapproved 2002), Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete. Appendix 1 (SI): Tables A1.5.3.3, A1.5.3.4(a), A1.5.3.6.
5. ACI Committee 301 — ACI 301-16, Specifications for Structural Concrete. §4.2.3.3.
6. ACI Committee 318 — ACI 318-19 / ACI 318M-19, Building Code Requirements for Structural Concrete. Tables 19.3.2.1 and 19.3.3.1; §26.4.3.1(b).
7. ACI Committee 209 — ACI 209R-92, Prediction of Creep, Shrinkage, and Temperature Effects in Concrete Structures. Eq. (2-1), Table 2.2.1.
8. ASTM C1074-19e1, Standard Practice for Estimating Concrete Strength by the Maturity Method.
9. ACI Committee 224 — ACI 224R-01 (Reapproved 2008), Control of Cracking in Concrete Structures, Table 4.1; ACI 224.1R-07, Causes, Evaluation, and Repair of Cracks in Concrete Structures.
10. The Concrete Society — Technical Report 22, Non-structural cracks in concrete, 4th ed., 2010.

11. ASTM C270, Standard Specification for Mortar for Unit Masonry — Table 1 (proportion specification), Table 2 (property specification); summarised in PCA, Masonry Mortars, 2004.
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13. AFNOR — NF EN 206-1 (April 2004), French National Annex: Table NA.F.1 and §NA.4.1 (values as reproduced by CIMbéton / Infociments, “La norme béton NF EN 206-1”, CT-T43, chap. 2). Later editions NF EN 206/CN (2014) and NF EN 206+A2/CN (2022) to be checked.
14. AFNOR — NF EN 1992-1-1/NA (2016), French National Annex to Eurocode 2 Part 1-1: Table 7.1NF and its notes.
15. ACI Committee 301 — ACI 301-16, Table 4.2.3.3.a (modification factor for standard deviation, interpolation between 15 and 30 tests).
16. ACI — Frequently Asked Questions no. 657, “air content and strength” (about 5 % strength reduction per 1 % air), concrete.org.
17. J. Bigun, G. Granlund — Optimal orientation detection of linear symmetry, Proc. 1st ICCV, 1987 (structure tensor; coherence).
18. BSI — BS 8666:2020 (bar shapes). EN 10080; BS 4449; ABNT NBR 7480; ASTM A615/A615M (bar sizes).
19. NOAA / National Weather Service — The Heat Index Equation (Rothfusz regression, NWS Technical Attachment SR 90-23, 1990) and Heat Index category chart.

10. About, privacy and support

CivilSuite collects no personal data: calculations, reports and photos stay on your device or in your own iCloud; the weather request sends the site position to Apple only. Privacy policy, terms of use and support: civilsuite.github.io/civilsuite.

Questions, errors, suggestions: civilsuite.app@proton.me or Settings › Send feedback.

Requirements: iOS / iPadOS 18.0 or later, or macOS 15.0 or later (native Mac app; one purchase covers iPhone, iPad and Mac with the same Apple Account). On iOS 18 to 25 and macOS 15 to 25, address search uses the Core Location geocoder; from iOS 26 and macOS 26, the MapKit one. On Mac, the camera is not offered (photos are imported from Photos or a file) and “Manage my subscription” opens the Subscriptions page of the Apple Account.

CivilSuite — calculation aids to be checked by a responsible engineer against the standards in force.